A Comprehensive Bayesian Multiple Linear Regression Analysis of the Survival Rates of Patients with Breast Cancer in the MSK-IMPACT Assay Dataset

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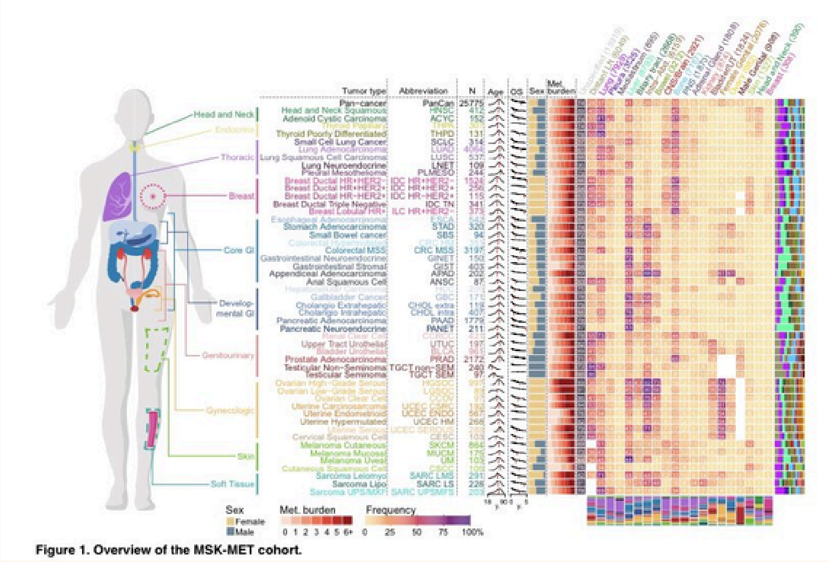
MSK Impact is a tumor sequencing method

Developed by Memorial Sloan Kettering to analyze genetic mutations

Contains data of 10,946 patients using the MSK-IMPACT assay.



2017 sequencing dataset is publicly available on the cBioPortal

cBioPortal
FOR CANCER GENOMICS

Our variables are clinical and genomic

Variable:	Type:	Range:
Cancer Type Detailed	categorical	
Number of Samples per Patient	integer	1-5
Fraction Genome Altered	float	0-1
Sex	categorical	Male/Female
Sample Type	categorical	Metastatic/Primary
DNA Input	integer	50-250
Matched Status	categorical	matched/unmatched
Metastatic Site	categorical	
Oncotree Code	categorical	
Overall Survival (Months)	float	0-40.4
Patient's Vital Status	categorical	DECEASED/ALIVE
Primary Tumor Site	categorical	
Sample Collection Source	categorical	Outside/In-House
Sample coverage	integer	48-2778
Somatic Status	categorical	Matched/Unmatched

Data Cleaning

1. Removed unnecessary variables
2. Filter down to the cancer type, Breast
3. Label encoding for binary variables
4. Dummy encoding for multi-level variables

Research Question

In what ways do our clinical and genomic variables influence the survival of patients from the MSK Impact 2017 cohort?

Methods: Weibull models survival times

$$T_i \mid \alpha, \beta_0, \boldsymbol{\beta}, \boldsymbol{x}_i \overset{ind}{\sim} \text{Weibull}(\alpha, (s_i))$$

$$s_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij}$$

T_i = observed survival time in months for patient i

s_i = patient-specific scale parameter based on clinical/genomic predictors

A larger s_i corresponds to longer expected survival time

Methods: Priors and Parameters

$$\beta_0 \sim \mathrm{Normal}(0, 1000)$$

$$\beta_j \sim \mathrm{Normal}(0, 2^2), \quad j = 1, \ldots, p$$

$$\alpha \sim \mathrm{Gamma}(1, 1)$$

$$\alpha < 1 : \text{decreasing hazard},$$
$$\alpha = 1 : \text{constant hazard},$$
$$\alpha > 1 : \text{increasing hazard}$$

Methods: Likelihood

observed survival/follow-up time in months

Weibull density, used when the patient died

Weibull survival probability, used when the patient was censored

event indicator

$$\delta_i = \begin{cases} 1, & \text{if death was observed} \\ 0, & \text{if the patient was censored} \end{cases}$$

$$L_i = f(t_i \mid \alpha, s_i)^{\delta_i} S(t_i \mid \alpha, s_i)^{1-\delta_i}$$

$$L(\alpha, \beta_0, \boldsymbol{\beta}) = \prod_{i=1}^{N} \left[f(t_i \mid \alpha, s_i)^{\delta_i} S(t_i \mid \alpha, s_i)^{1-\delta_i} \right]$$

Methods: Posterior

$$p(\alpha, \beta_0, \boldsymbol{\beta} \mid \mathbf{t}, \boldsymbol{\delta}, X) \propto L(\alpha, \beta_0, \boldsymbol{\beta}) \cdot p(\beta_0) \cdot p(\boldsymbol{\beta}) \cdot p(\alpha)$$

$$p(\alpha, \beta_0, \boldsymbol{\beta} \mid \mathbf{t}, \boldsymbol{\delta}, X) \propto \prod_{i=1}^{N} f(t_i \mid \alpha, s_i)^{\delta_i} S(t_i \mid \alpha, s_i)^{1-\delta_i} \cdot p(\beta_0) \cdot p(\boldsymbol{\beta}) \cdot p(\alpha)$$

Methods

- Exploring the most influential variables
- JAGS to simulate our distribution and derive posterior summaries
- Making diagnostic plots to verify the convergence of the model
- Using Bayesian Hypothesis Testing to explore the relationship between the influential variables and the target
- Performing posterior predictive check to evaluate the model fit

Importance of Variables

	abs_effect
beta[1]	1.1222704710
beta[20]	0.6577032711
beta[16]	0.6456427346
beta[19]	0.5642668742
beta[4]	0.4081329784

Cancer type detailed breast invasive cancer

Specimen preservation type FFPE

Sex

Specimen preservation type DNA

Cancer type detailed breast invasive lobular carcinoma

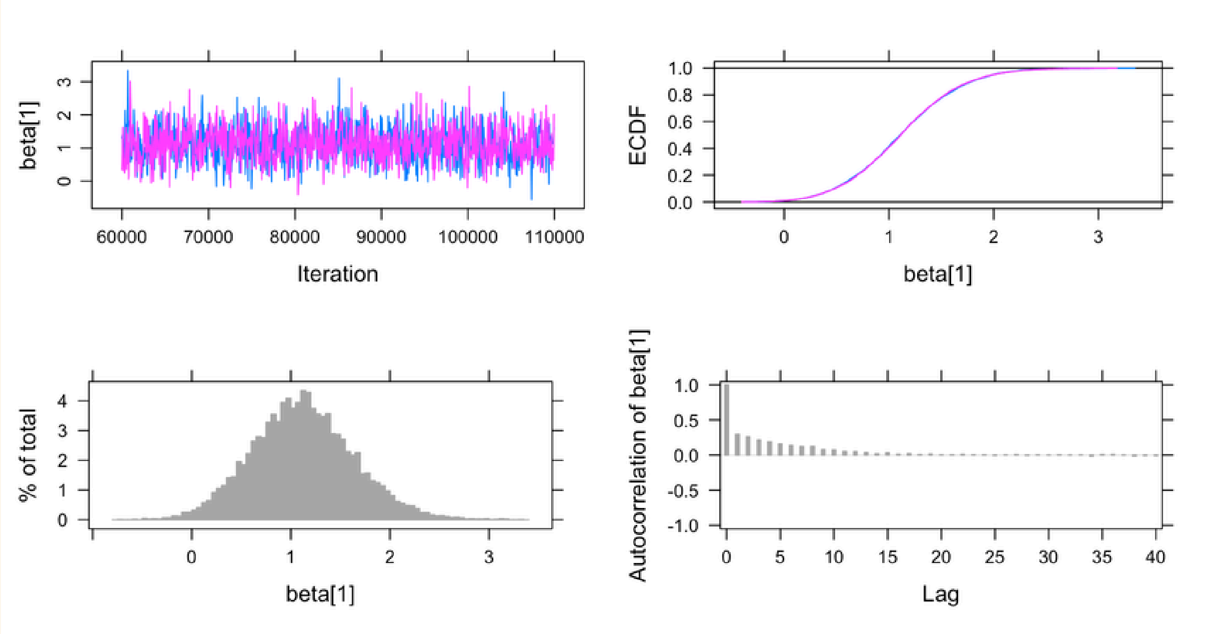
JAGS Posterior Summary

beta	predictor	posterior_mean	posterior_median	CI_lower	CI_upper
beta[1]	Cancer.Type.DetailedBreast Invasive Cancer, NOS	1.1222704710	1.110887884	0.15943601	2.18259228
beta[20]	Specimen.Preservation.TypeFFPE	-0.6577032711	-0.593418341	-2.92902838	1.36162555
beta[16]	SexMale	-0.6456427346	-0.671012157	-1.31847127	0.17570927
beta[19]	Specimen.Preservation.TypeDNA	-0.5642668742	-0.506113963	-2.86807465	1.46185766
beta[4]	Cancer.Type.DetailedBreast Invasive Lobular Carcinoma	0.4081329784	0.422944949	-0.33075878	1.06238529

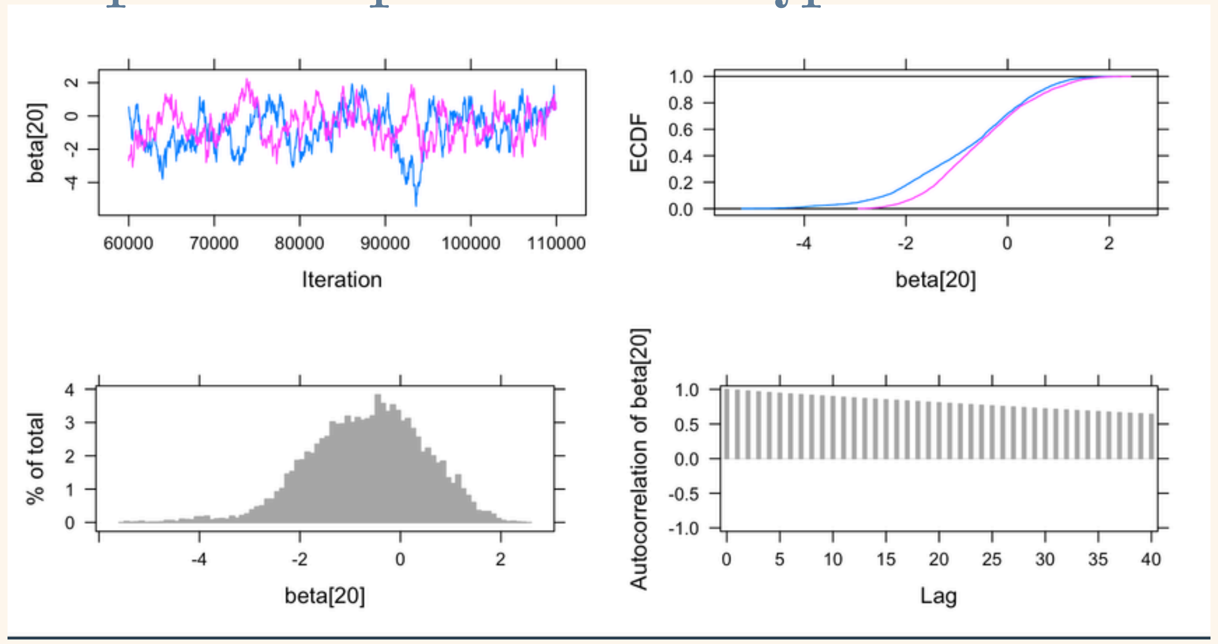
	P_positive	P_negative	abs_effect	abs_weight
beta[1]	0.9893	0.0107	1.1222704710	1.1222704710
beta[20]	0.2941	0.7059	0.6577032711	0.6577032711
beta[16]	0.0506	0.9494	0.6456427346	0.6456427346
beta[19]	0.3233	0.6767	0.5642668742	0.5642668742
beta[4]	0.8766	0.1234	0.4081329784	0.4081329784

Diagnostic Plots of Important Variables

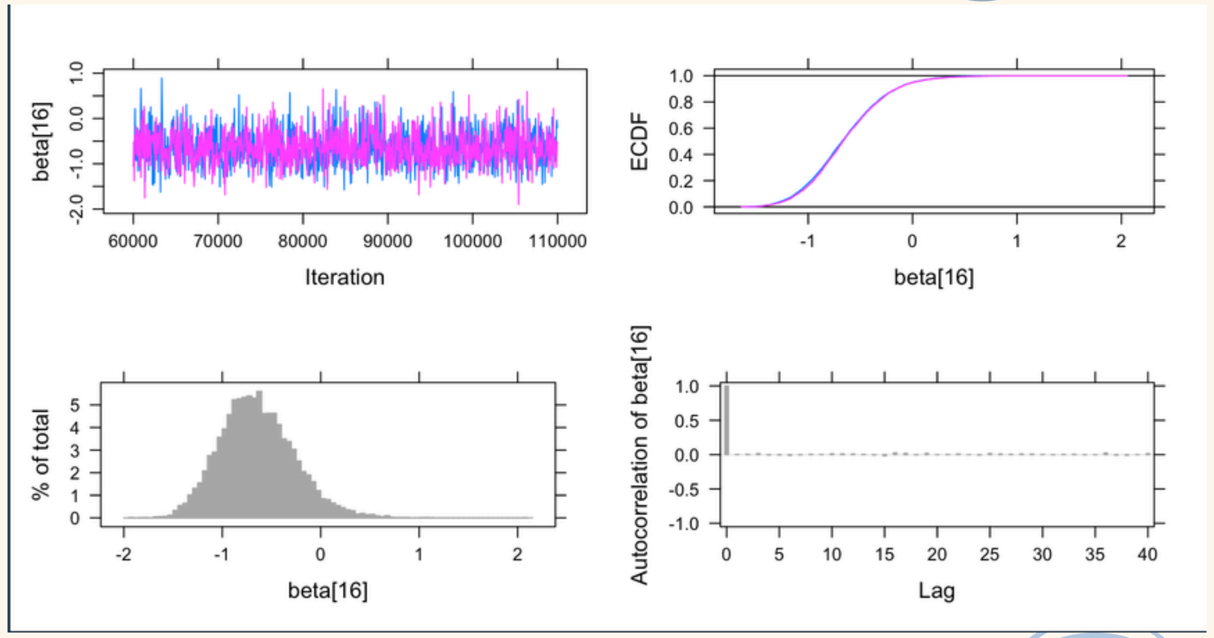
Cancer type detailed breast invasive cancer



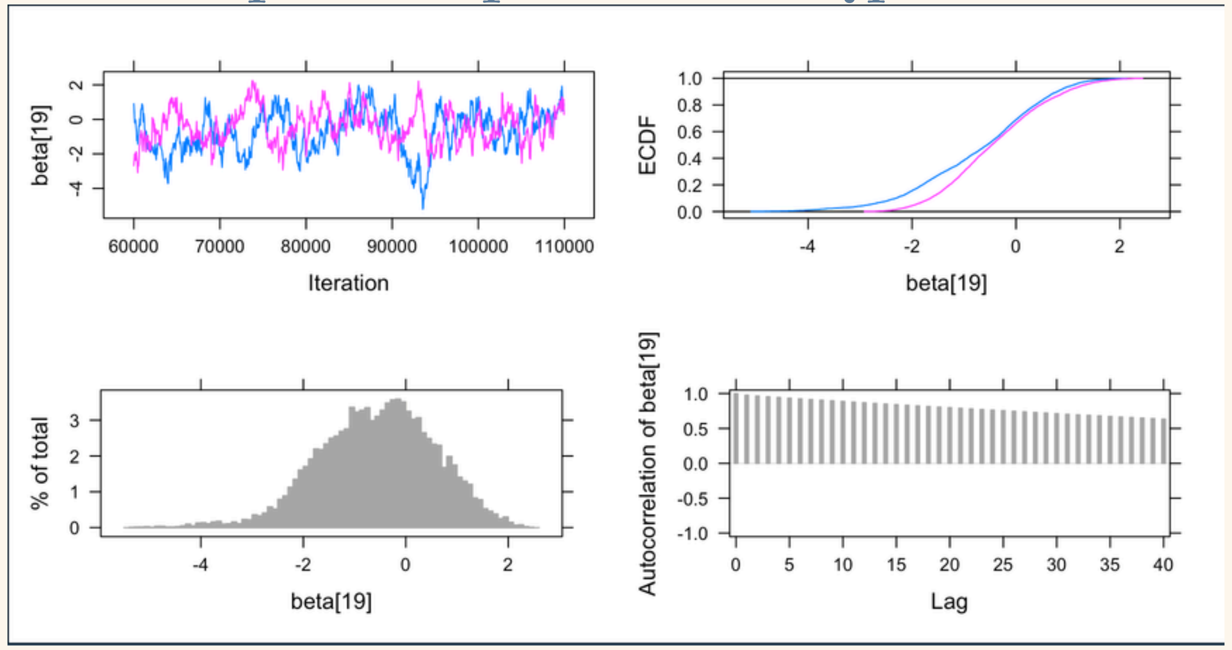
Specimen preservation type FFPE



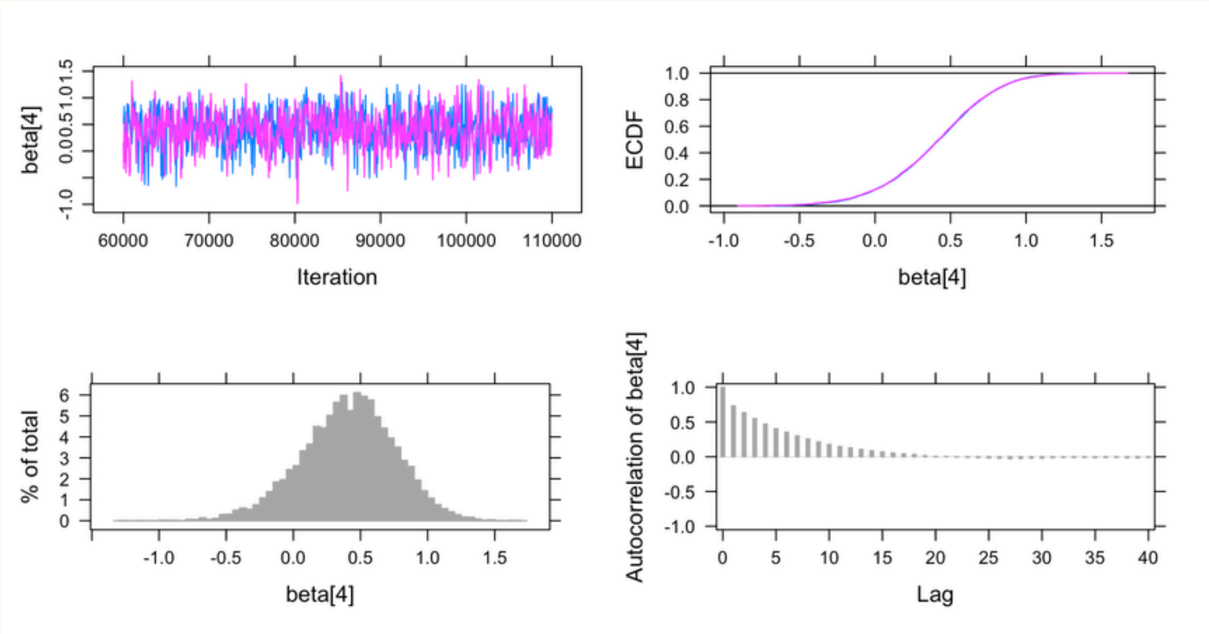
Sex



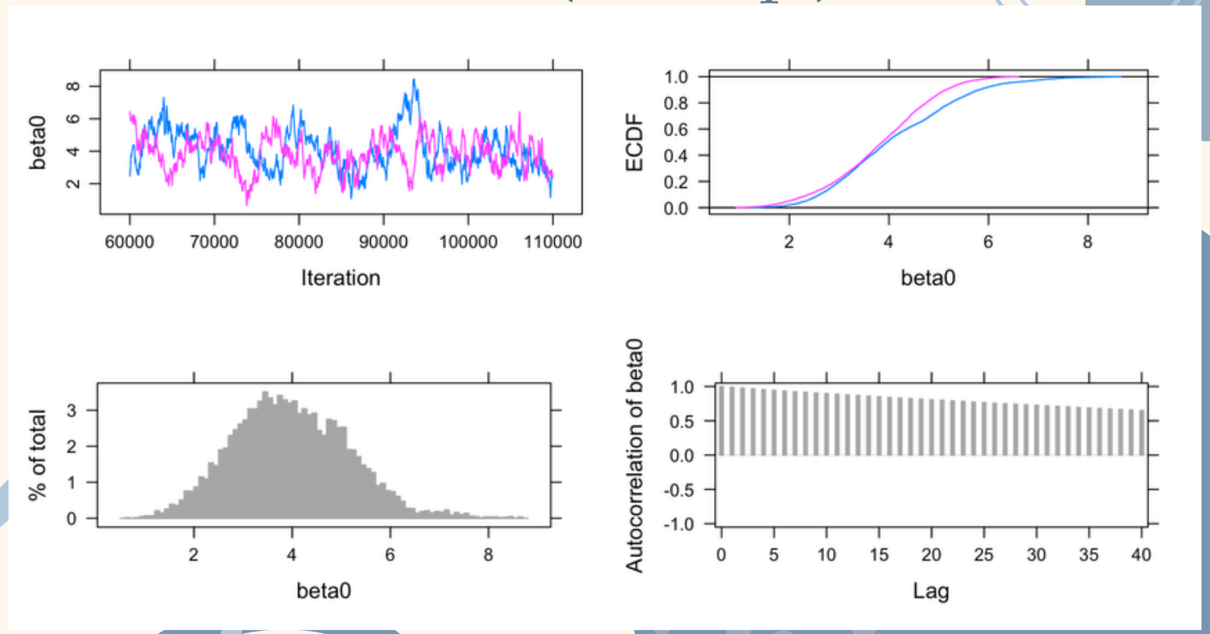
Specimen preservation type DNA



Cancer type detailed breast invasive lobular carcinoma



beta0 (intercept)



Bayesian Hypothesis Testing

Variable name	probability pos	probability neg
Cancer Type Detailed Breast Invasive Cancer	0.9893	0.0107
Specimen Preservation Type FFPE	0.2941	0.7059
Sex	0.0506	0.9494
Specimen Preservation Type DNA	0.3233	0.6767
Cancer Type Detailed Breast Invasive Lobular	0.8766	0.1234

Very high association with longer survival times

Relative association with shorter survival times

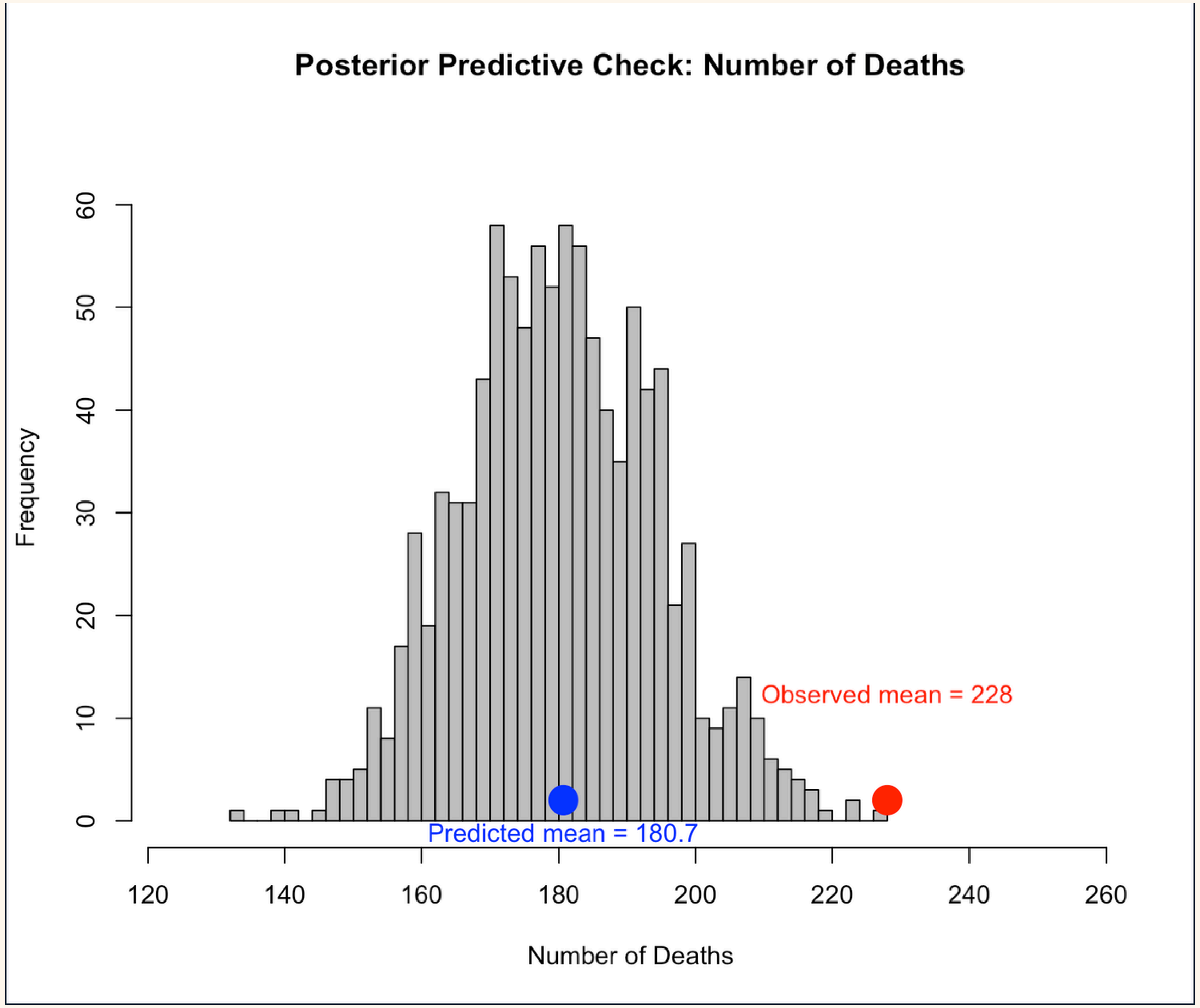
Male patients had much higher associations with shorter survival times

Weak association with shorter survival times

High association with longer survival times

Note: Male patients were underrepresented in our dataset, so no concrete conclusions about this variable can be made

Posterior Predictive Check



The model underpredicted the number of observed deaths, suggesting that it overestimated survival. This may be because most observations were censored, meaning the model only knows those patients survived past their last follow-up time. Additionally, the Weibull model assumes a relatively simple hazard structure, which may be too restrictive for the true survival pattern in the data.

References

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“Weibull Distribution.” Wikipedia, 19 Oct. 2020, en.wikipedia.org/wiki/Weibull_distribution.